

Q2 Library Book Management [BL3 – Apply]

10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
CLASS Book
```

```
    PRIVATE title : STRING
```

```
    PRIVATE author : STRING
```

```
    PRIVATE isbn : STRING
```

```
    PRIVATE isAvailable : BOOLEAN
```

```
    PUBLIC METHOD setDetails(t, a, i)
```

```
        title <- t
```

```
        author <- a
```

```
        isbn <- i
```

 Save Answer

Your Answer

```
author <- a
isbn <- i
isAvailable <- TRUE
END METHOD

PUBLIC METHOD issueBook( )
  IF isAvailable = TRUE
THEN
    isAvailable <- FALSE
    OUTPUT "Book Issued: ", title
  ELSE
    OUTPUT "Book Not Available: ", title
  END IF
END METHOD

PUBLIC METHOD returnBook( )
  IF isAvailable = FALSE
THEN
    isAvailable <- TRUE
    OUTPUT "Book Returned: ", title
```

 Save Answer

Your Answer

```
        OUTPUT "Book Returned: ", title
    ELSE
        OUTPUT "Book Already Available: ", title
    END IF
END METHOD

PUBLIC METHOD getTitle( )
    RETURN title
END METHOD

PUBLIC METHOD getAuthor( )
    RETURN author
END METHOD

PUBLIC METHOD getISBN( )
    RETURN isbn
END METHOD

PUBLIC METHOD getAvailability( )
    RETURN isAvailable
END METHOD

PUBLIC METHOD displayDetails( )
    OUTPUT "Title: ", getTitle( )
```

END METHOD displayDetails()

OUTPUT "Title: ", getTitle()

OUTPUT "Author: ", getAuthor()

OUTPUT "ISBN: ", getISBN()

OUTPUT "Available: ", getAvailability()

END METHOD

END CLASS

BEGIN

DECLARE b1, b2, b3 AS Book

CALL b1.setDetails("Java Basics", "Author A ", "111")

CALL b2.setDetails("OOP Concepts", "Author B", "222")

CALL b3.setDetails("Data Structures", "Author C", "333")

CALL b1.issueBook()

CALL b1.returnBook()

CALL b2.issueBook()

CALL b2.issueBook()

CALL b3.displayDetails()

END